Photonics



Open Optical Solutions for AI Scaling

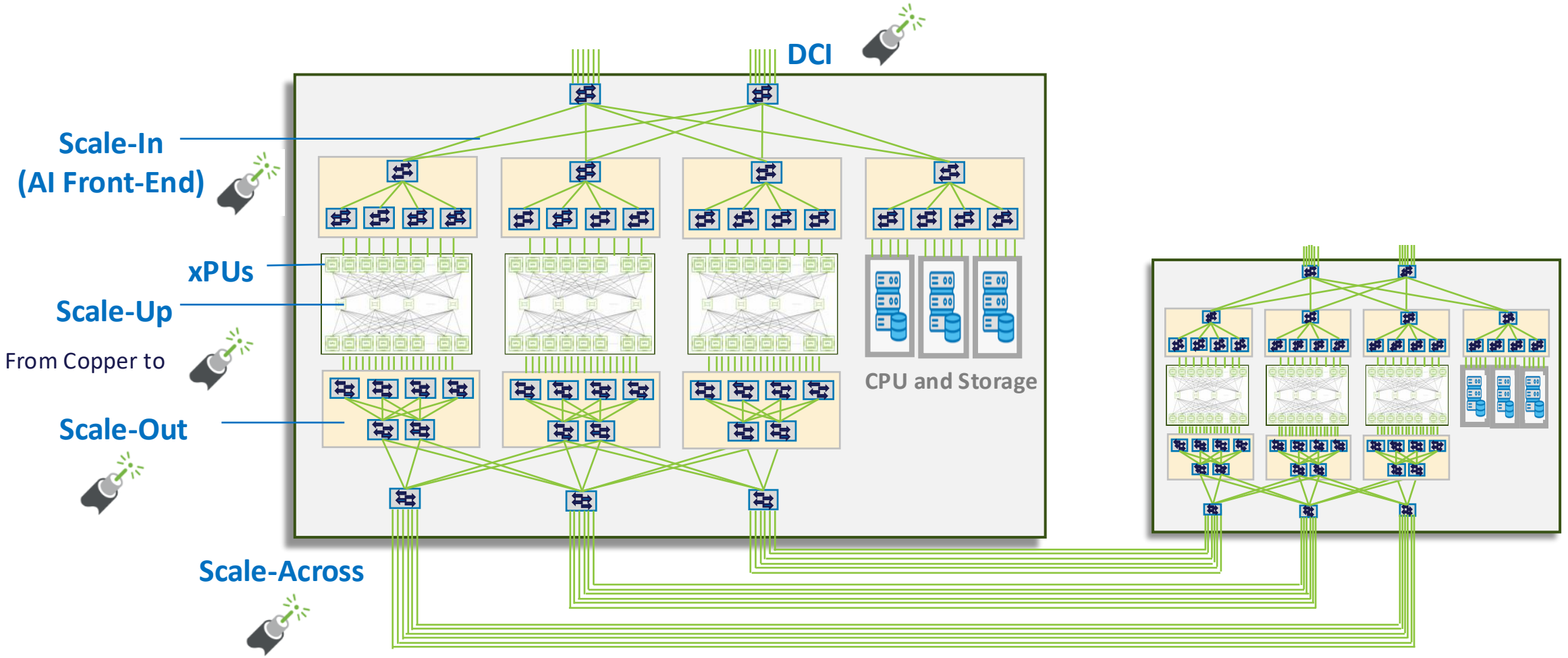
Rang-Chen (Ryan) Yu

VP Marketing, TeraHop

OCP SILVER Member

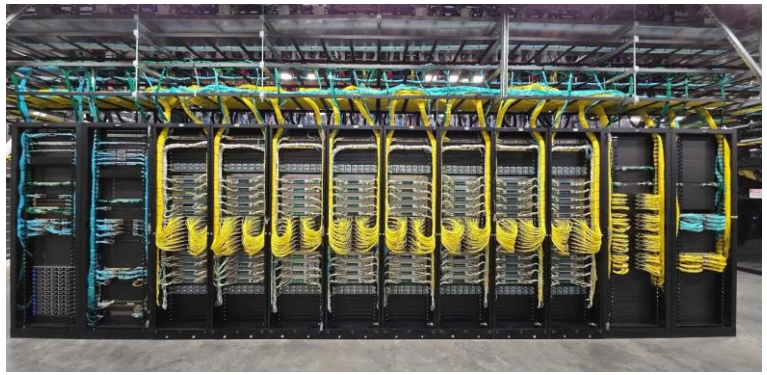
AI Scaling Networking

- AI Scaling requires complex networking with scale up, scale out, scale across and scale-in
- Optical connectivity solutions to meet AI Scaling Challenges



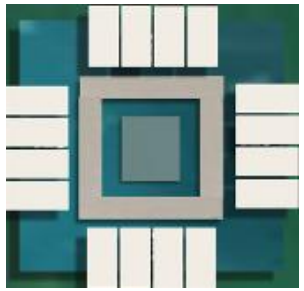
AI DC Scaling Challenges

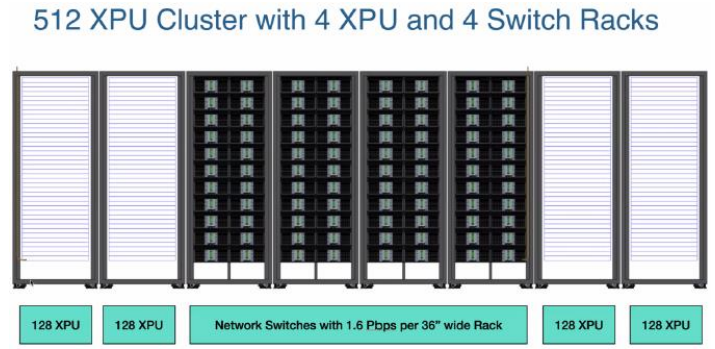
BW Capacity challenge: AI scale up BW ~ 10x more than scale out, ~ 10-20Tbps/xPU



Source: Nvidia

Density:
ASIC shoreline density, System faceplate density





Source: Andy @ Arista illustrating OSFP challenges

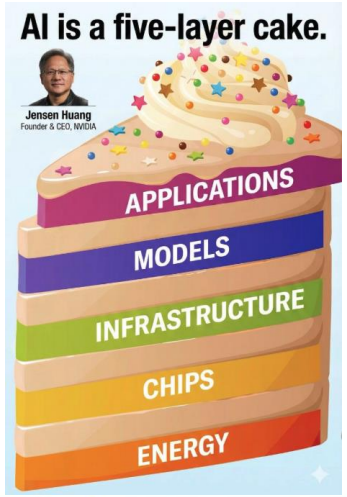
Fiber: Intra-DC >10M fiber, exceeding fiber tray capacity; Inter-DC > 4M fiber, growing

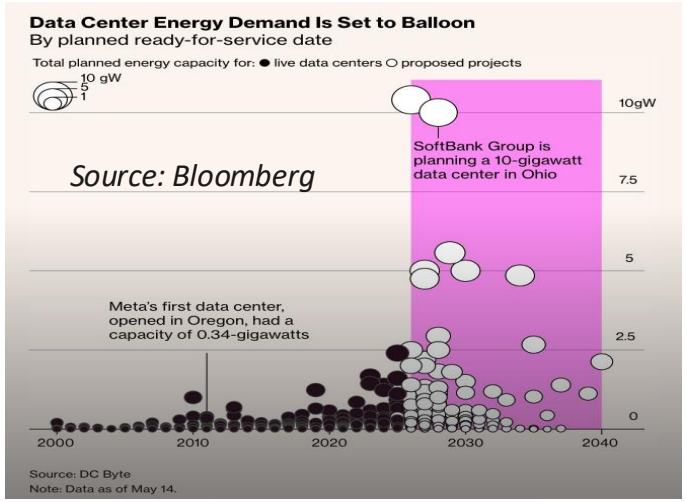




Source: Rob Stone, Meta

Power: AI DC foundation is Energy..

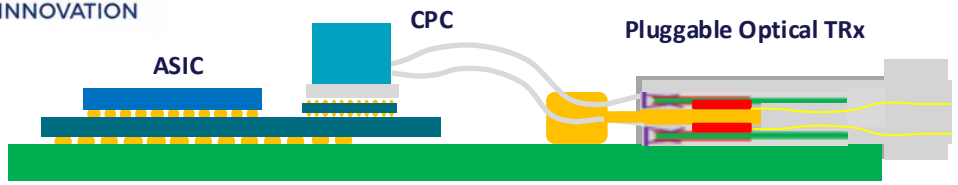




Open Optical Solutions for AI Scaling

XPO OPTICS INNOVATION

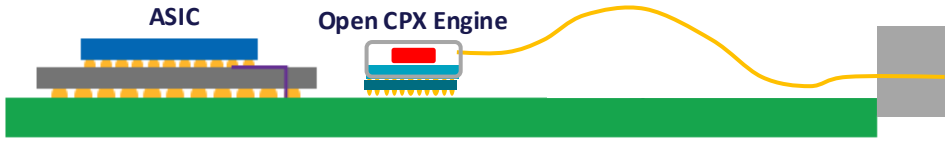
12.8T XPO MSA: Solutions for BW Capacity and system Faceplate Density





OPEN CPX MSA

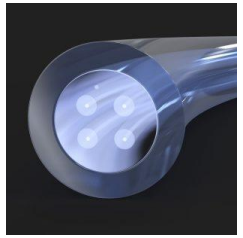
6.4T Open CPX MSA: Solutions for low power and high density

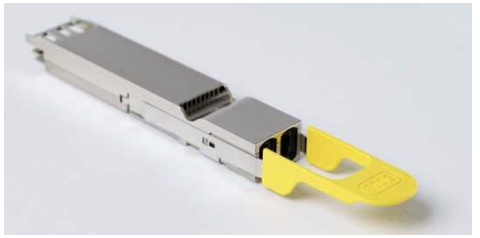




SDM4 MCF MSA

SDM4 MCF MSA: Solutions for fiber infra. challenges





XPO MSA Addressing Capacity/Density Challenges

XPO
OPTICS INNOVATION

- 8 x 1.6T Capacity, 4 x Faceplate Density
- 1st Pluggable with Integrated liquid cooling
- Supports wide range of optics and copper
  - Architectures: FRO, RTLR, LPO
  - Optical PMDs: DRn, FRn, OCI, Coherent-Lite, ZR..
- Wide industry support: 150+ companies joined MSA

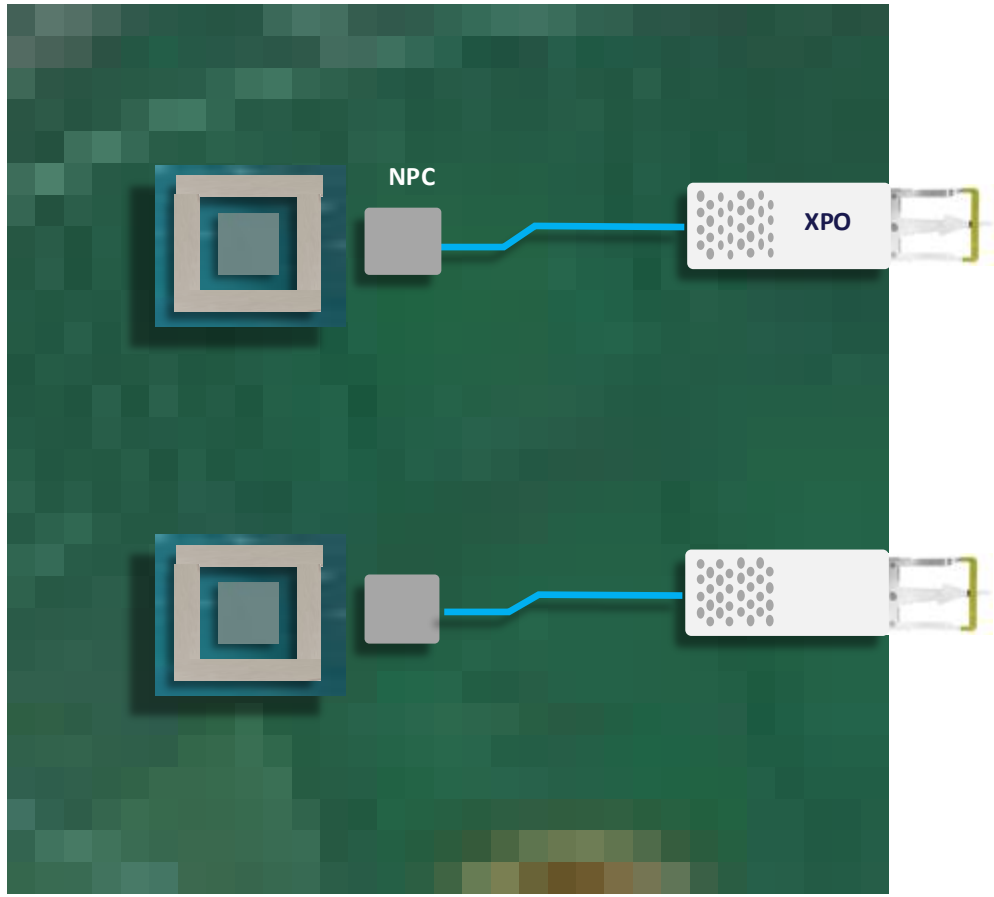




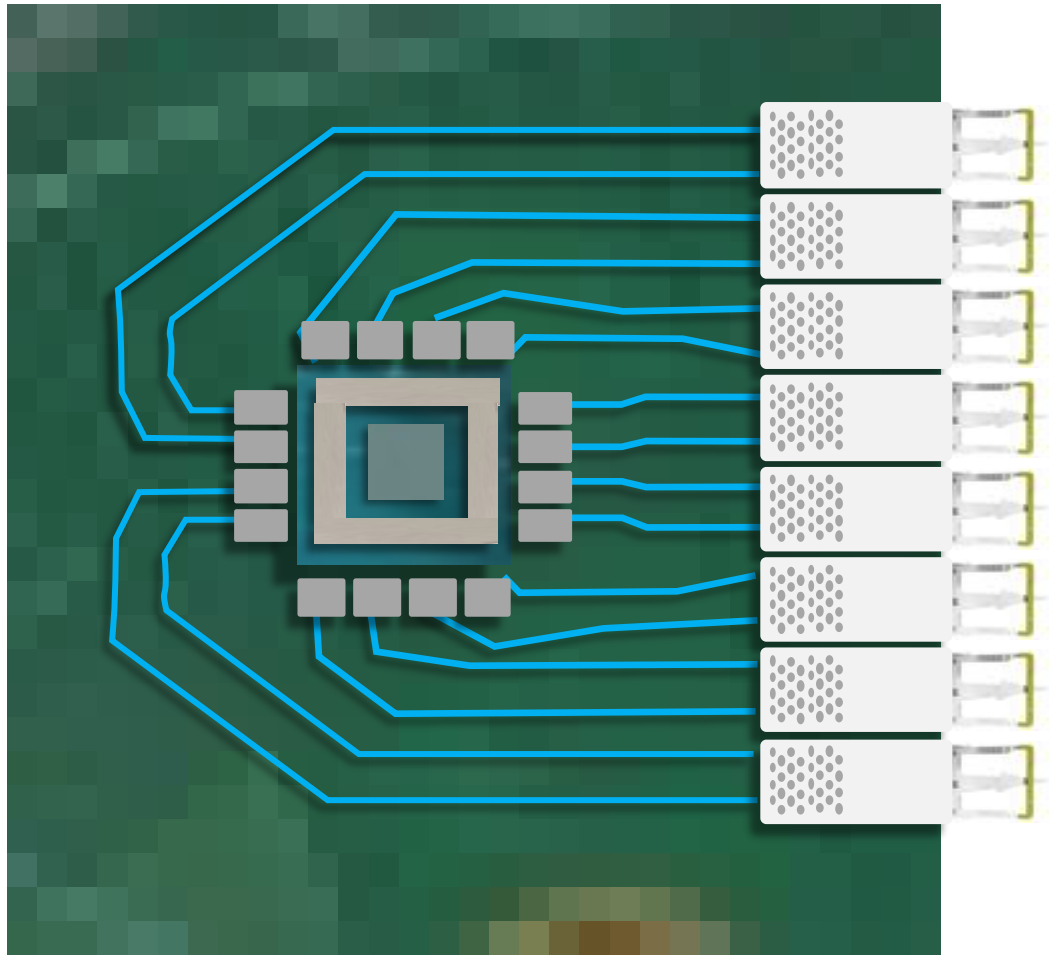
Use Cases of XPO

XPO
OPTICS INNOVATION

xPU optical interface for Scale Up



High-Capacity Switch Interface for Scale Up/Scale Out



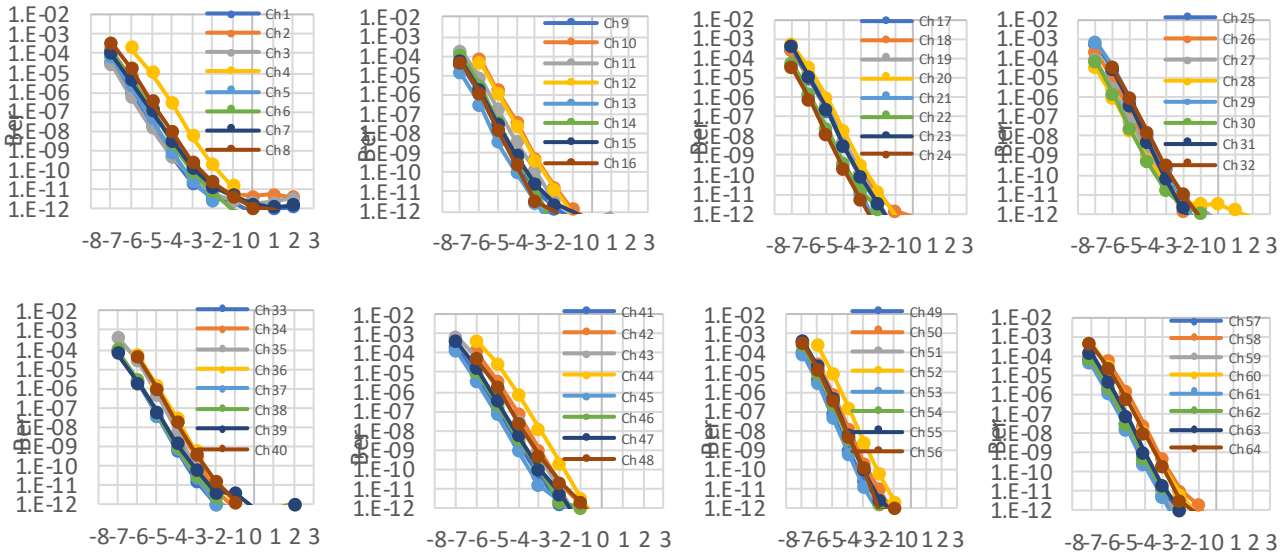
Industry 1st 12.8T XPO Performance

XPO OPTICS INNOVATION

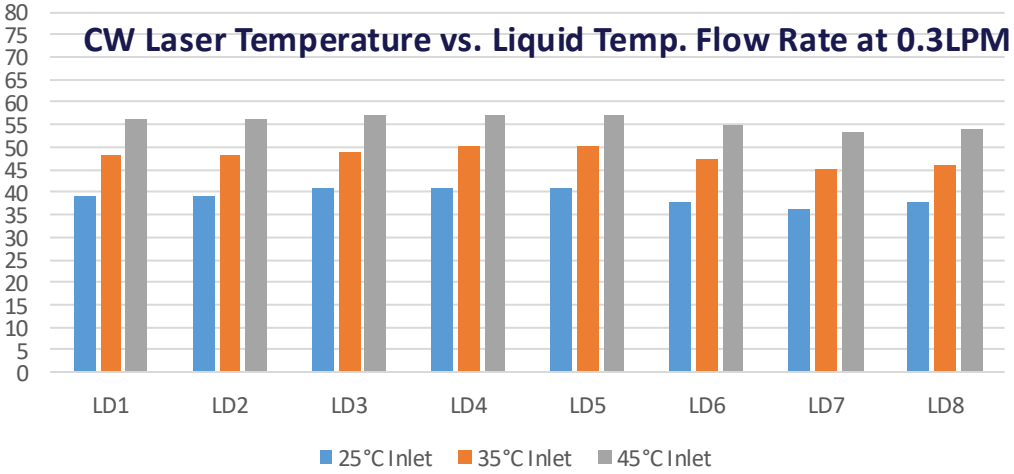
64 x Transmitters



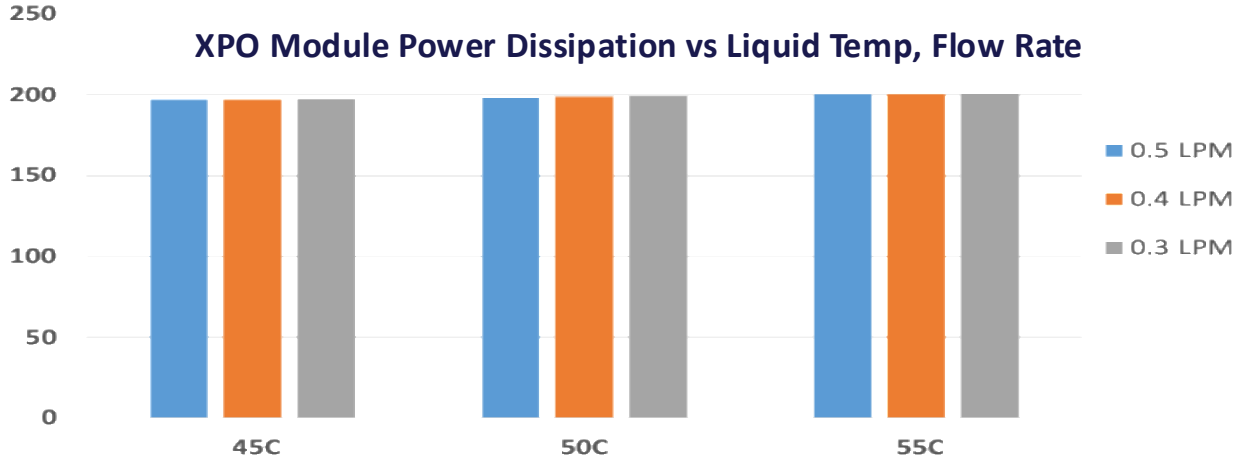
8 x 1.6T-DR8 Link Pre-FEC BER



CW Laser Temperature vs. Liquid Temp. Flow Rate at 0.3LPM



XPO Module Power Dissipation vs Liquid Temp, Flow Rate



Open CPX Addressing Density/Power Challenges

OPEN CPX MSA

Open CPX MSA

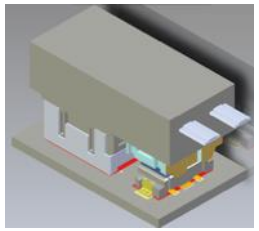
Specification for

Co-packaged and Near-Package Connector and Optics Module

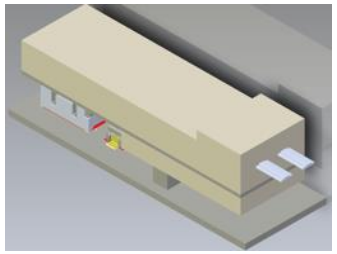
Rev 1.0

2026 September 16

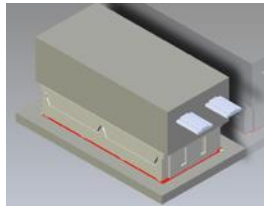
Type 1 Connector, ELM



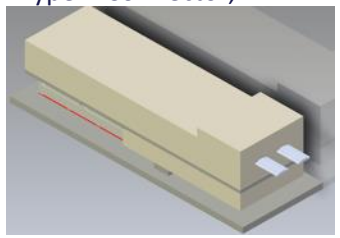
Type 1 Connector, ILM



Type 2 Connector, ELM



Type 2 Connector, ILM



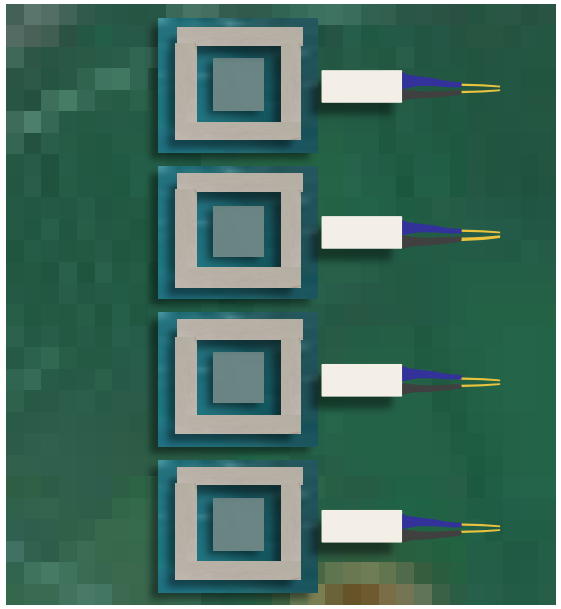
The Open CPX MSA founding members include Ciena, Coherent, Marvell, Molex, Samtec, and TeraHop (alphabet order).

The Open CPX MSA early Contributor members include Accton Technology, Intel, Nexthop Systems, Qualcomm and TE Connectivity Solutions.

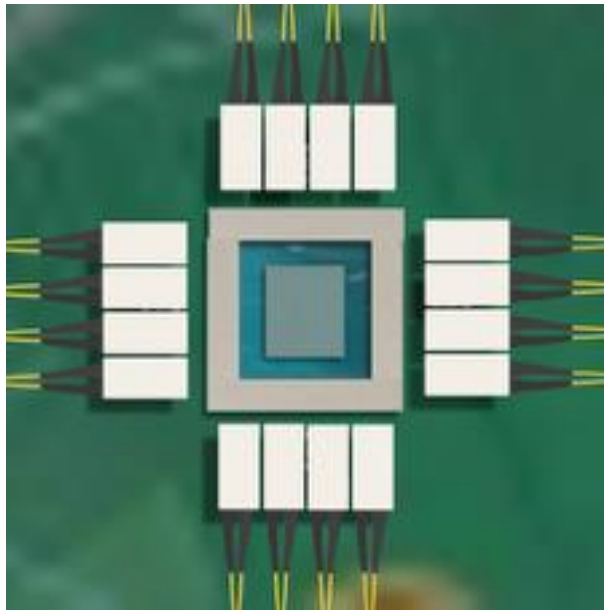
Use Case Examples of Open CPX Optics

OPEN CPX MSA

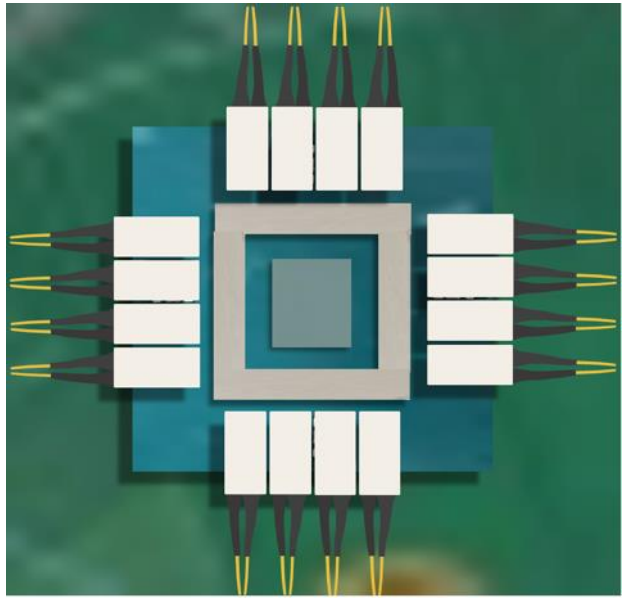
NPO for xPU Scale up



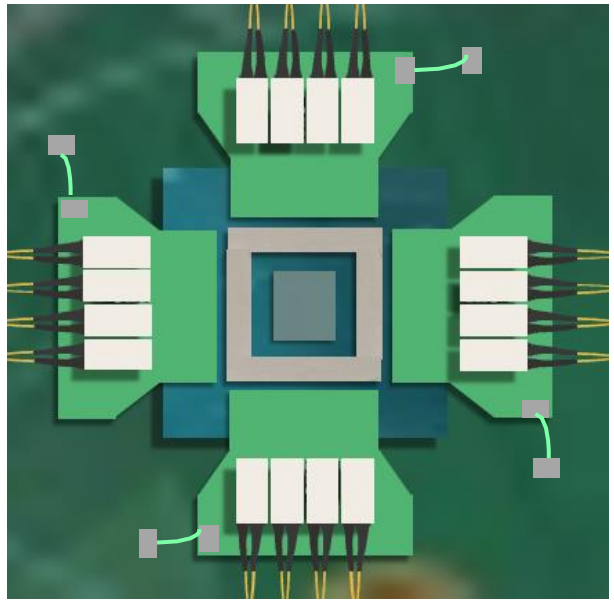
NPO for Switch Scale up/Out



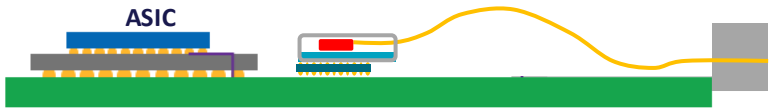
CPO for Switch Scale up/Out



IOA (Integrated Optical Attachment)



Open CPX NPO



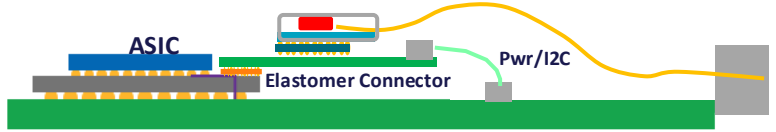
- Optical Engine on Host PCB
- Type 1 or Type 2 Connectors

Open CPX CPO



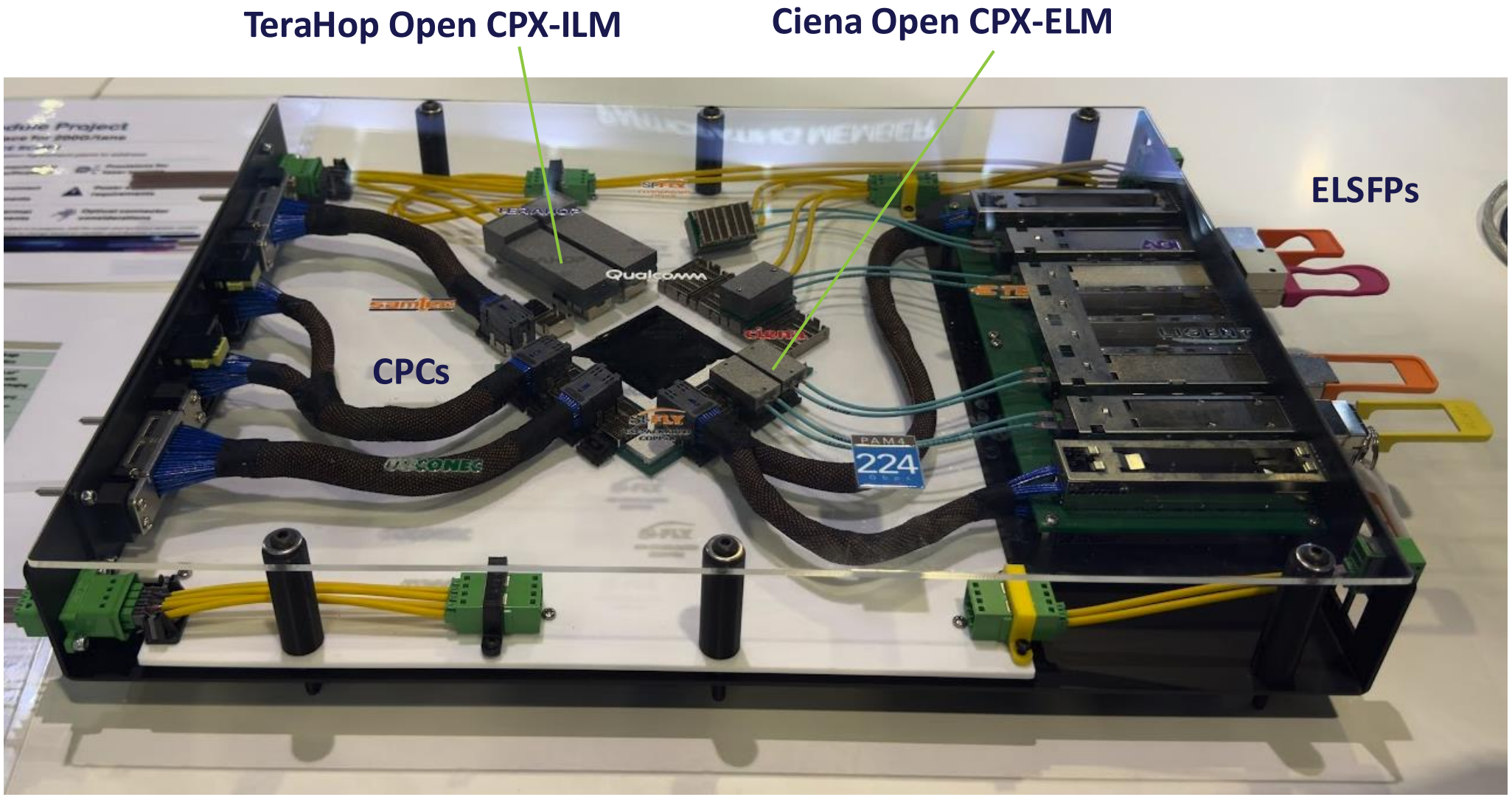
- Optical Engine on ASIC substrate
- Type 1 connector with high speed DP on ASIC substrate, power/I2C on host PCB

Open CPX IOA



- Optical engine on HDI board attach to ASIC substrate via elastomer high density, high speed low loss connector
- Type 1 or type 2 connectors

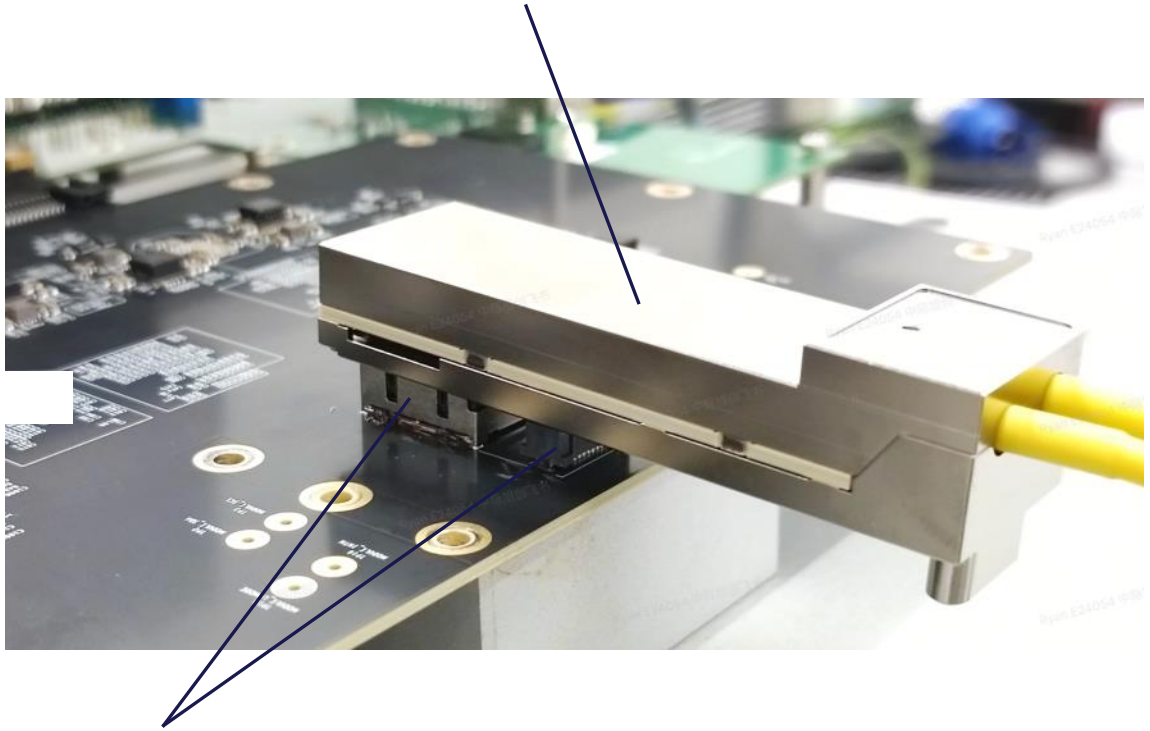
ECOC OIF Show Case of Open CPX Ecosystem



TeraHop “Diablo-1” 6.4T Open CPX Engine

OPEN CPX MSA

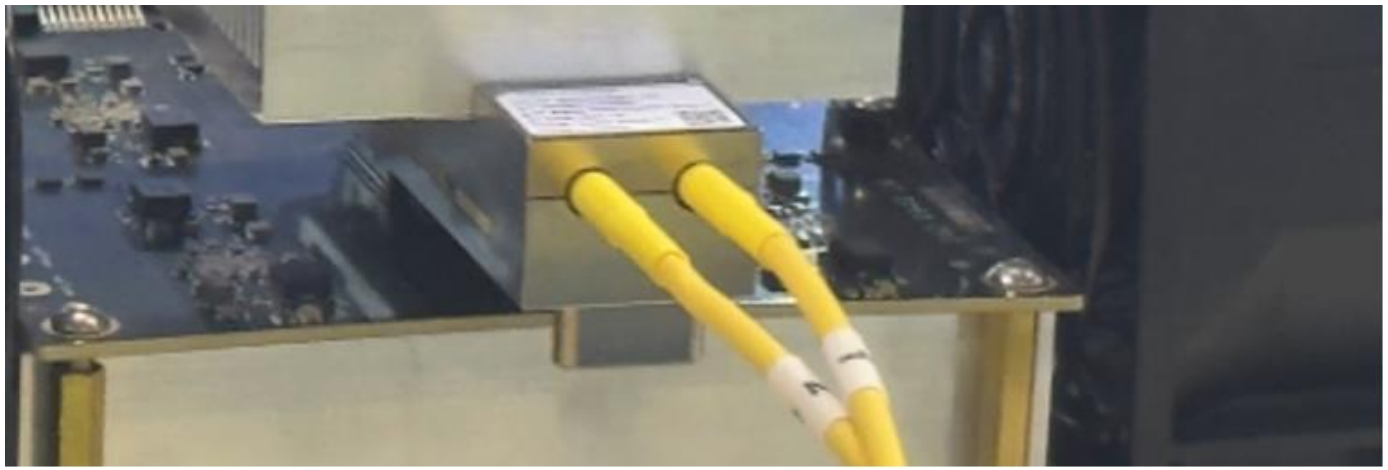
Silicon Photonics Optical Engine with Integrated laser Source

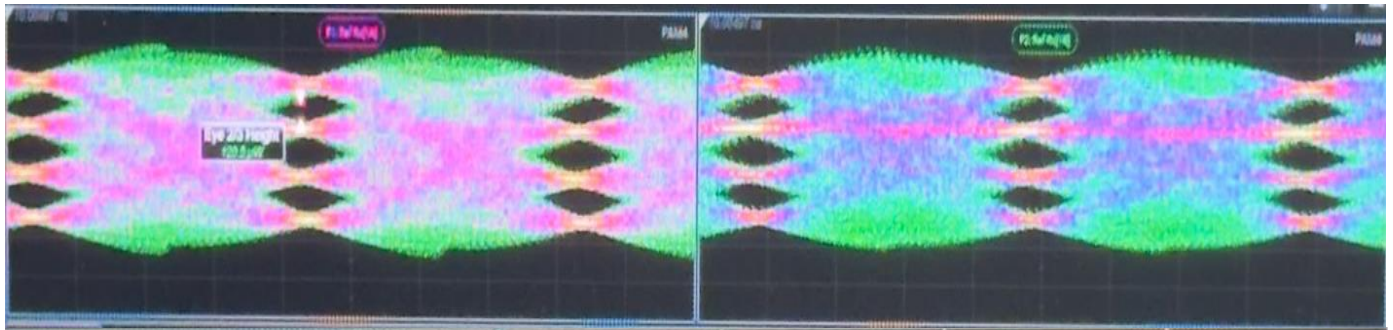


Open CPX Type 1 Connectors

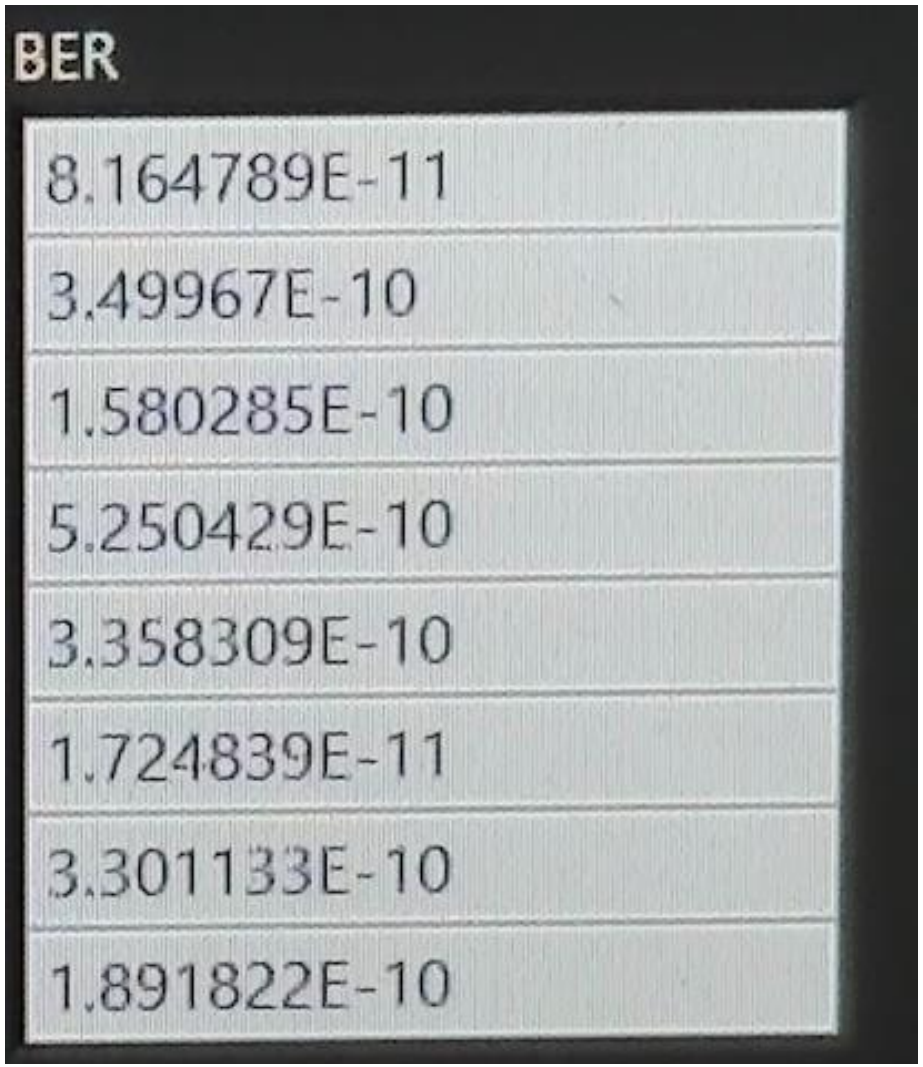


TeraHop “Diablo-1” 6.4T NPO ECOC OIF Demo

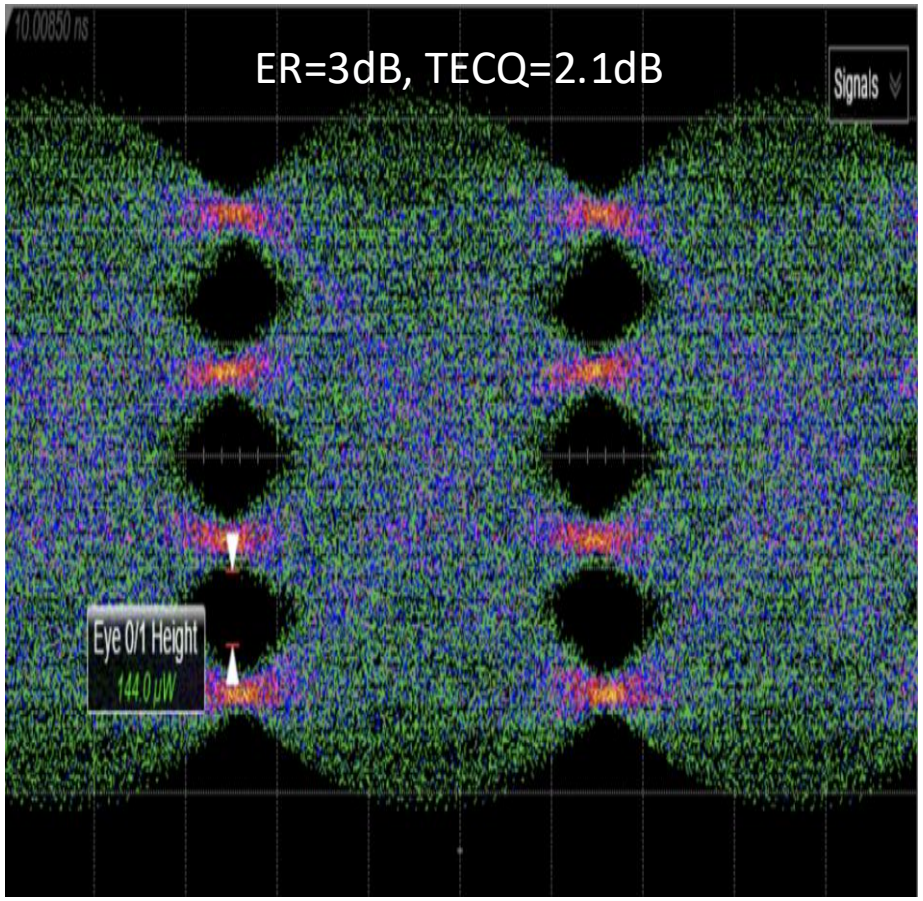


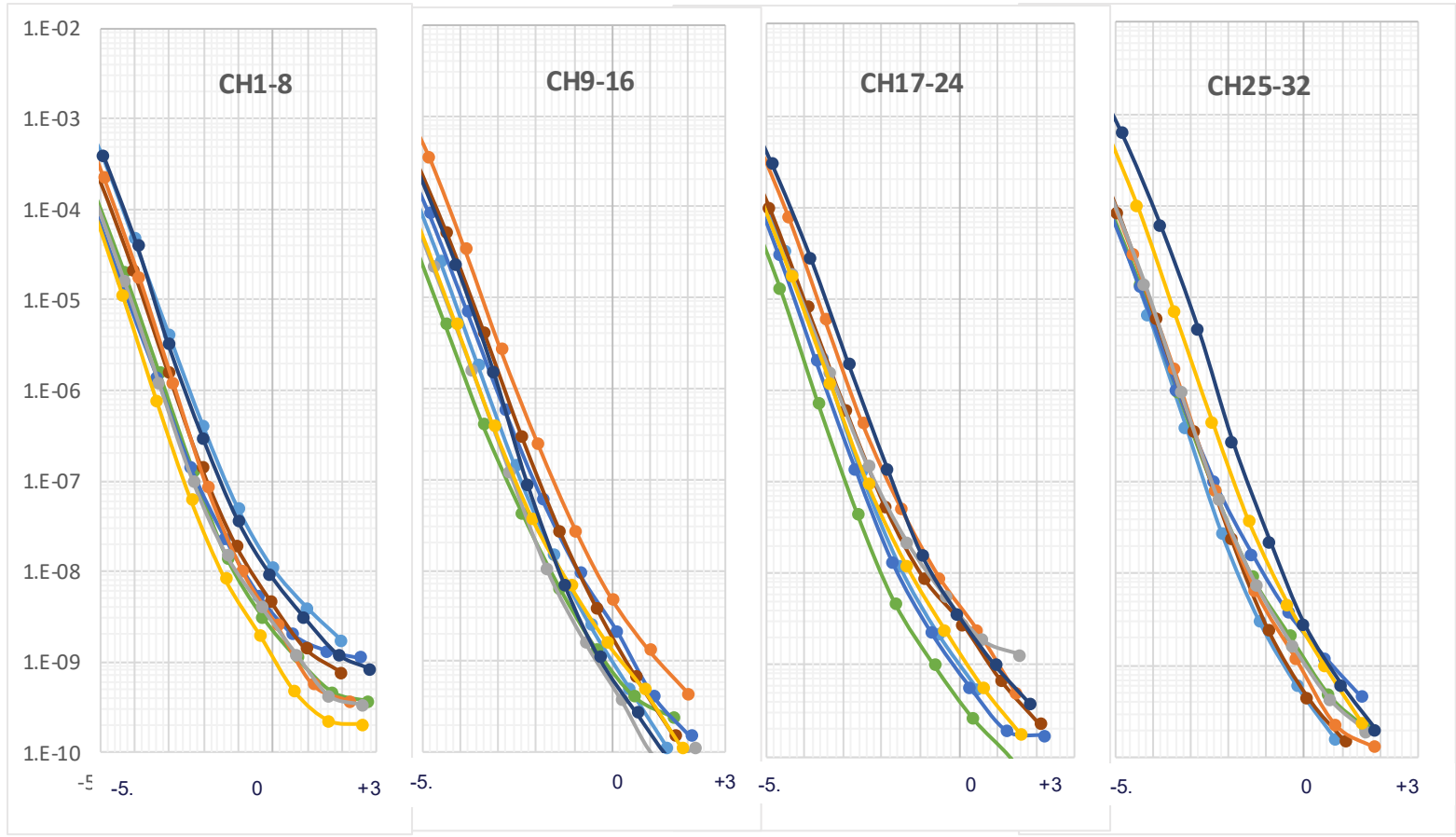


Measurement		Current	Minimum	Maximum	Count
Outer ER	F2	2.936 dB	2.936 dB	3.019 dB	402
Outer ER	F1	3.105 dB	3.104 dB	3.111 dB	402
TDECQ	F1	2.43 dB	2.42 dB	2.45 dB	402
TDECQ	F2	2.48 dB	2.27 dB	2.48 dB	402



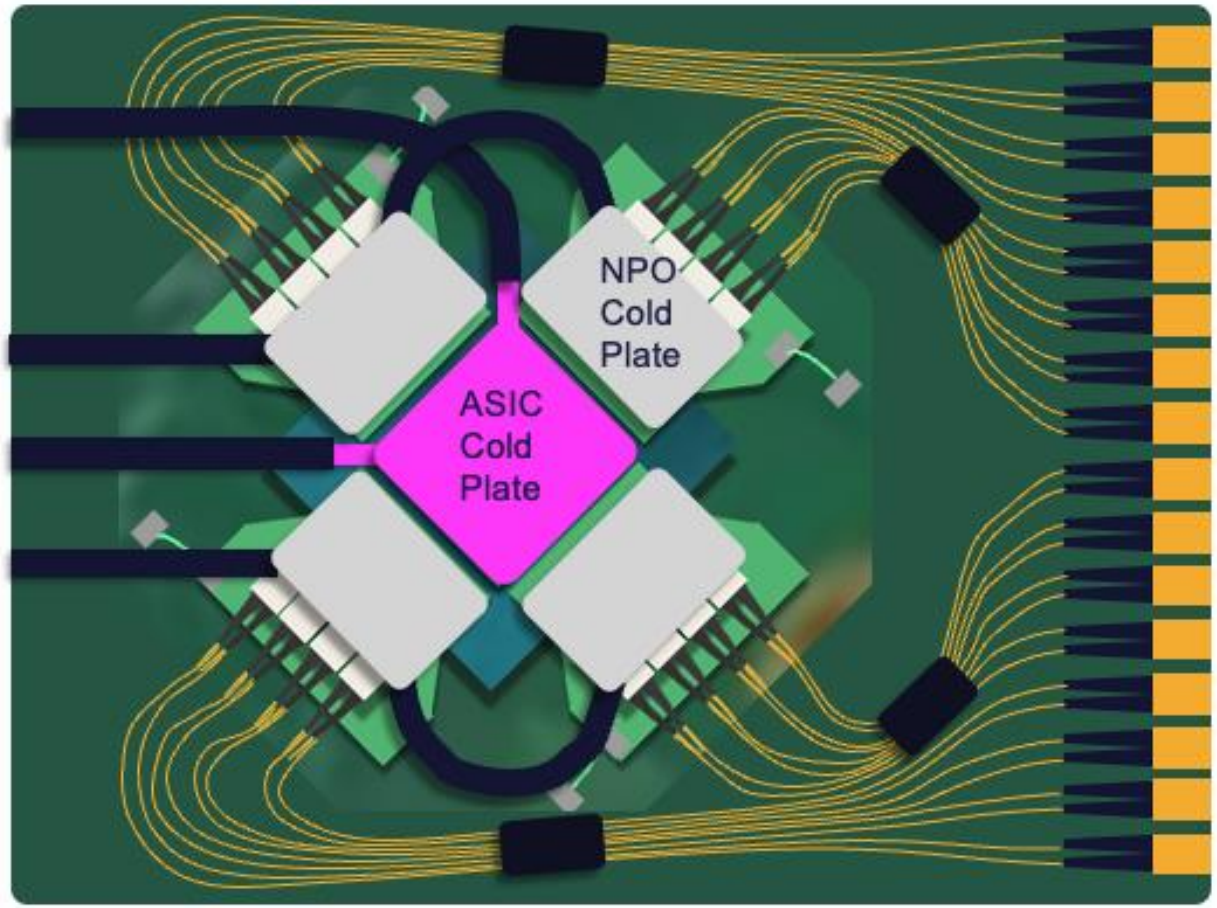
TeraHop “Diablo-1” 6.4T NPO Prelim. Test Data





* Insertion loss between ASIC to NPO ~ 12dB

Open CPX NPO with Integrated Laser



Integrated laser with proven low FIT

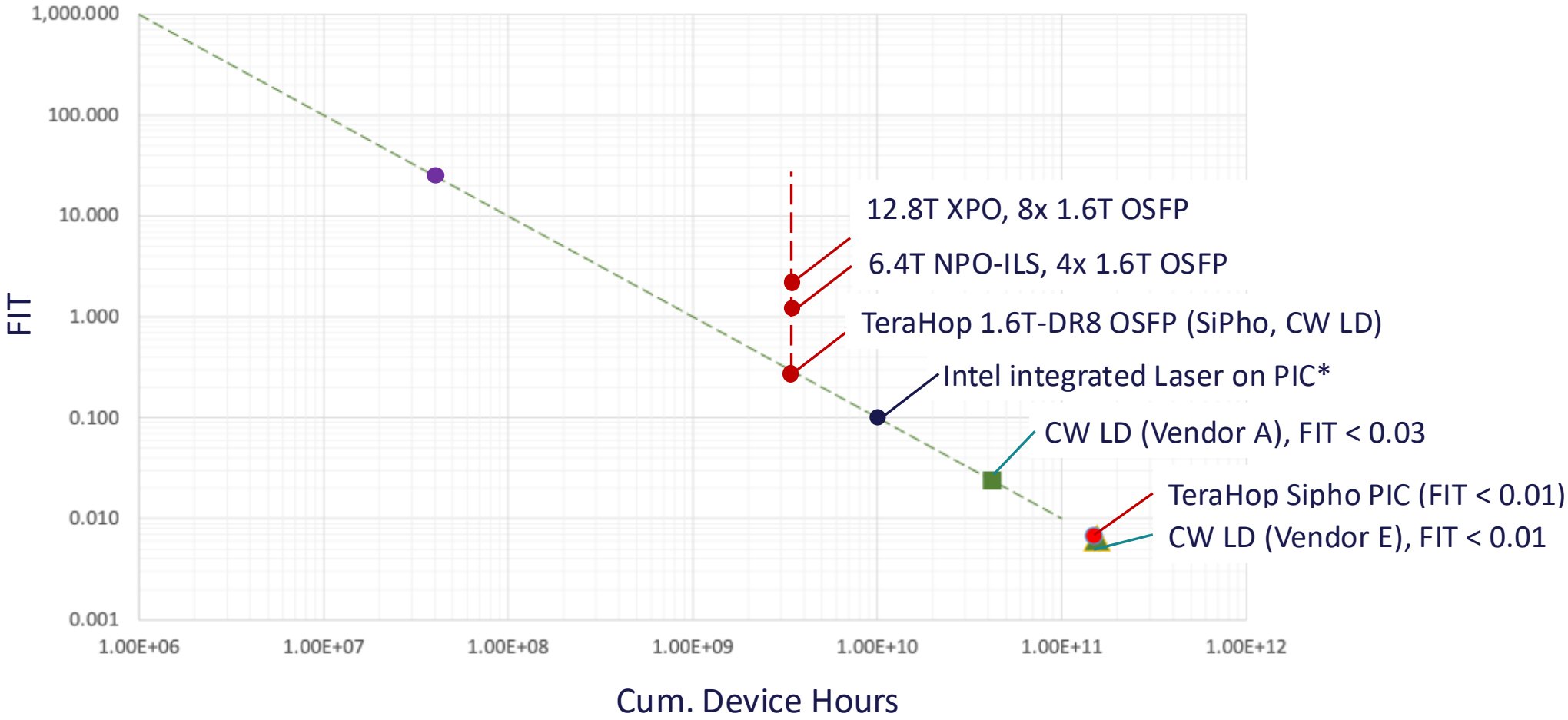
- NPO close to, but separate from ASIC
  - Separate cold plate from ASIC
  - Critical components including LD at modest temperature
- Low power ~ 5.5pj/b, including internal laser
- Simple SMF pigtail termination to front panel
  - No extra connector loss and potential failure

ELSFP Disadvantage

- lose >3dB power from extra coupling and connectors
- →requires > 2x CW LD optical power (200-300mW+)
  - lacking field reliability data
  - increase CW LD power dissipation by ~ 4x
- PM fiber adding additional power loss and cost
- Blind-mate connector adding loss and failure
- Higher cost, power and less reliable

XPO and Open CPX Engine Reliability Discussion

- TeraHop shipped > 25M Silicon Photonics optical transceivers, with PIC >100B Device-Hours, FIT< 0.01



* Source: Intel. https://www.intel.com/content/www/us/en/newsroom/news/intel-unveils-first-integrated-optical-io-chiplet.html

SDM4 MCF MSA Addressing Fiber Challenges



AFL

SUMITOMO ELECTRIC

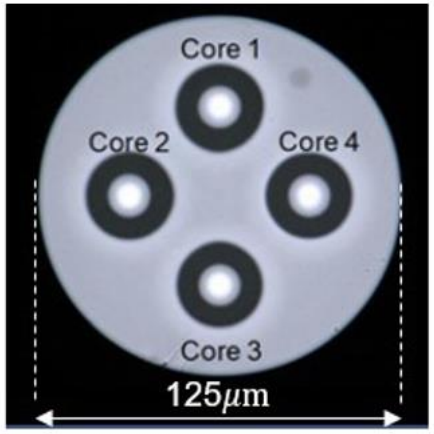
CORNING

TERAHOP

- **SDM4 MCF MSA** Launched (3/11/2026)
- **Founding members**: AFL, Corning, Sumitomo, TeraHop
- **Goals**: Define common geometrical, optical, and mechanical requirements to ensure multi-vendor interoperability.
- **Applications**: Optimized for short-reach, high-density O-band links in AI data centers and intra-campus networks
- **Release 1.0** published on 9/17/2026

Multi-Core Fiber for AI DC

SDM4 MCF MSA



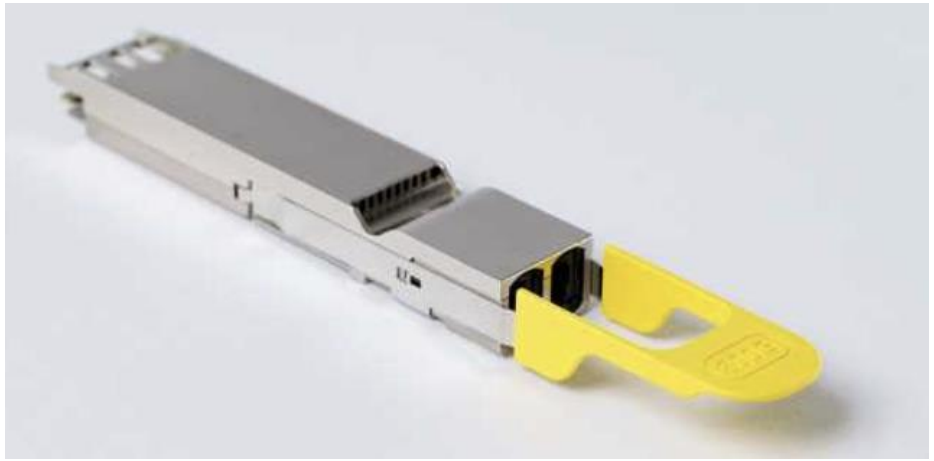
- Multi-core fiber: transmit multiple channels of signals over physically separated cores in 1 fiber
- Each core compatible with the conventional SMFs
- 2x2 4-Cores MCF are actively considered for AI Data center deployment

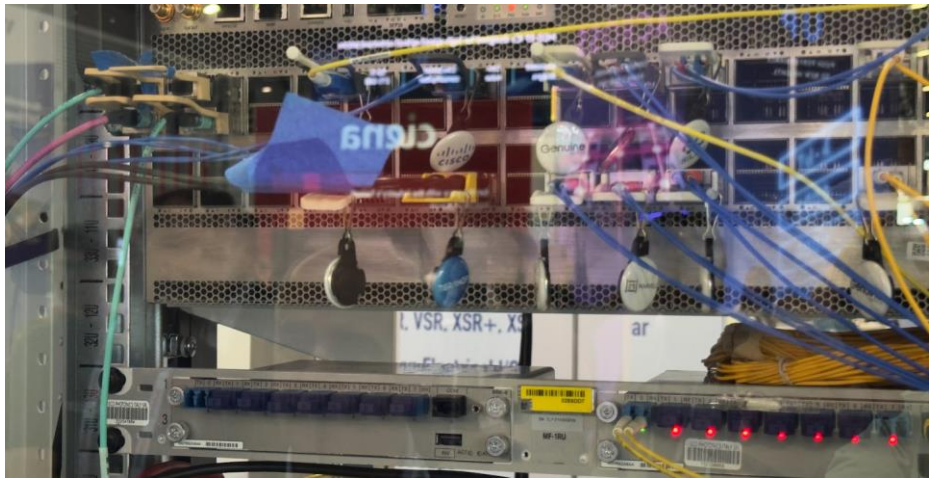


Source: Corning, OFC 2026 Demo

- **Key Benefits:**
  - **Fiber count reduction**
    - Reduce size and weight of optical fiber cables
    - Reduction of fiber handling complexity
    - Reduce fiber interface terminations by 4x
    - Reduce pathway required (cable trays inside DC, underground duct/conduits,..)
      →Reduce construction/installation labor and time
      → faster AI capacity Deployment

MCF Native Optical Transceivers Demo





ECOC 2026 OIF Demo with TeraHop 800G-MCF



OFC 2026 demo with TeraHop 800G-MCF transceiver and Corning MCF

Impact of SDM4 MCF to AI DC Fiber Infra

Normal SMF





Source: Meta

From Normal SMF
To SDM4-MCF





Conceptual fiber reduction with SDM4-MCF, generated by Google Gemini

www.ethernetallian

Call to Action

- AI scaling drive demanding optical interconnect solutions
  - BW capacity, density, power, fiber infra..
- Open optical solutions keeping up with Innovation
- Please join open optics community:
  - XPO MSA: https://xpomsa.com/
  - Open CPX MSA: https://www.opencpxmsa.org/
  - SDM4 MCF MSA:https://terahop.com/newsinfo/150.html







Thank You!

